



FoodGPT

LOYIHA MA'LUMOTNOMASI · TECHNICAL DOSSIER

AI orqali «bugun nima yeyman?» savolini hal qiluvchi platforma

O'zbekiston restoran discovery + bron infratuzilmasi.
Tabiiy tilda so'ra → AI yaqin, ochiq, mos joyni topadi
→ bron, navigatsiya, taxi bir oqimda.

Monorepo · 4 loyiha

Next.js 15

Gemini AI

Voice (STT/TTS + realtime)

uz · ru · en

Website ☒ tayyor

FoodGPT nima

Yadro g'oya, muammo va strategik pozitsiya.

FoodGPT — «AI restoran qidiruvi» EMAS. Yadro qiymat va himoya (moat) — restoranlarning **real-time data + bron infratuzilmasi** (OpenTable yo'li). AI — eshik, data — uy.

 Muammo

O'zbekistonda odam och qolganda qayerda, nima, aynan hozir tayyorligini tez topa olmaydi. Barcha restoranlarni birlashtirgan platforma yo'q. Odam Google Maps'ga kirib ovqat izlamaydi — shu bo'shliq.

 Yechim

«Shashlik yemoqchiman» yoki «200 ming budgetga oilaviy joy» deb yoz/gapir → AI 30 soniyada mos joyni topadi, bron qiladi, yo'l ko'rsatadi. Suhbat, tavsiya, qaror — menyu ko'rish emas.

Delivery modeli — strategik farq

- **Zakaz FoodGPT'da:** AI chatdan menyu → savat → to'lov. Bu yadro — foydalanuvchi tajribasi va data shu yerda.
- **Dastavka hamkor orqali:** yetkazishni FoodGPT o'zi qilmaydi — restoranning o'z kuryeri yoki hamkor kuryer API (Yandex/Uzum). Yengil operatsiya, og'ir logistikasiz.
- **Xulosa:** «yana bir delivery ilova» emas. Discovery + zakaz bizniki, yetkazish — hamkor infratuzilma. Moat baribir **data + bron**da.

Repo tuzilishi — monorepo

Bitta repo, to'rtta mustaqil loyiha (uch mijoz + bitta backend). Root'da turbo/nx yo'q — oddiy papkalar.

LOYIHA	NIMA	STEK	HOLAT
website/	Marketing sayt: g'oya, muammo/yechim, funksiyalar, demo, aloqa	Next.js 15	 TAYYOR
backend/	Umumiy API: Gemini AI, voice (STT/TTS+realtime), geo-qidiruv, restoran DB	Node + Fastify	 PLAN
tg-mini-app/	Telegram ichida ishlaydigan asosiy iste'molchi ilova	Vite + React 18	 PLAN
pwa/	O'rnatiladigan veb-ilova (offline, push, home-screen)	Next.js 15 + Serwist	 PLAN

Mijoz → backend: `tg-mini-app` va `pwa` bir xil `backend` API'ni ishlatadi — bitta AI/voice/geo mantiqi, ikki (keyin uch) mijoz. `website` — statik marketing, backend'siz. Har loyiha o'z `package.json`, dev serveri bilan mustaqil (`cd website && npm run dev`).

Yadro oqim: AI · voice · geo

tg-mini-app va pwa da bir xil. Kalitlar — faqat backend'da.

Foydalanuvchi

- | (1) lokatsiyani yoqadi → koordinatalar (lat/lng)
- | (2) so'raydi: matn YOKI ovoz ("shashlik yemoqchiman")



BACKEND (umumiy API server)

- | VOICE LAYER (abstract, provayder ENV bilan almashadi):
 - Primary: real-time speech-to-speech (Gemini Live)
 - Fallback: STT → Gemini (matn) → TTS
- | AI (Gemini): niyat (taom, budjet, kayfiyat, kompaniya, vaqt)
- | GEO-QIDIRUV: koordinata + niyat → yaqin, ochiq, mos joylar
- | RESTORAN DATA: real-time holat, menyu, narx, bron



Mijoz: yaqin joylar ro'yxati (masofa/narx/holat)
 → bron | ZAKAZ (menyu→savat→to'lov) | navigatsiya | taxi
 + AI javobini ovozda o'qiydi (TTS/realtime)



Qat'iy qoida — kalitlar

Gemini/voice/xarita API kalitlari **FAQAT backend'da**.
 Hech qachon mijoz bundle'iga (TG/PWA/website) tushmaydi. Mijoz faqat backend endpoint'iga so'rov yuboradi.



Voice — abstract qatlam

Bitta interfeys; ichida provayder almashadi.
 VOICE_PROVIDER=realtime|pipeline. Token/limit tugasa fallback'ga o'tadi — matn baribir ishlaydi (voice = qulaylik, gracefully degrade).



Geo

Lokatsiya mijozda olinadi (TG location, PWA geolocation), backend'ga koordinata yuboriladi.
 Saralash (haversine masofa + holat) backend'da.



Til

AI javoblari uz/ru/en — foydalanuvchi tiliga mos. O'zbek birinchi. System prompt lokal kontekstni biladi (osh/norin/tandir, budjet so'mda).

website/ — marketing sayt ✓ TAYYOR

Yagona ishlaydigan kod. Info-only landing — investor/hamkor uchun birinchi taassurot.

Framework	Next.js 15.1.6 (App Router) + React 19 + TypeScript strict
Styling	Tailwind CSS 3.4 + CSS custom properties (design tokenlar)
i18n	next-intl 3.26 — uz (default) · ru · en · locale prefix as-needed
Theme	next-themes — light/dark, tizim + toggle, token darajasida
Animatsiya / Ikon	Framer Motion 11 · lucide-react · Inter font (self-host)
Deploy	Railway (NIXPACKS) — root package.json → website/ build+start
Kod hajmi	13 ta site-bo'lim komponenti (~1200 qator TSX) · i18n ~33 KB (3 til)

Sahifa bo'limlari (13 ta, tartib bilan)

#	BO'LIM	VAZIFA
1	Hero	Sarlavha + aylanadigan placeholder input (typewriter)
2	ProblemSolution	Muammo → yechim qarama-qarshiligi
3	Stats	Bozor/traksiya raqamlari (tabular-nums)
4	Features	Asosiy imkoniyatlar gridi
5	Tech	Texnologik yadro (AI/voice/geo)
6	HowItWorks	Oqim: so'ra → top → bron
7	UseCases	Foydalanish stsenariylari
8	ForRestaurants	B2B taklif (restoranlar uchun)
9	DemoVideo	Demo video (placeholder)
10	Launch	Ishga tushirish / roadmap
11	Team	Jamoa
12	Contact	Aloqa
13	Footer	Havolalar + til/tema

Nav + Hero'da mock.ts namunaviy restoran data (Besh Qozon, Choyxona №1, Afsona Grill...) va matchRestaurants() keyword-router demo uchun ishlatilgan.

Design tizimi — barcha loyihalarda bir xil

Estetika: Apple × ChatGPT × Airbnb × Linear. Warm minimalism. Etalon: `website/globals.css`.

HIS: iliq · aqlli · ishonchli · premium · mazali. **QOCHISH:** delivery-marketplace koʻrinishi (Yandex Eats), chegirma bannerlari, qizil ustunligi, neon AI dashboard, ortiqcha animatsiya.

Rang tokenlari (light — warm cream, pure white EMAS)

TOKEN	QIYMAT	ROL
 <code>--bg</code>	<code>#fbf7f2</code>	Iliq krem fon (orange tomon mikro-bias)
 <code>--surface</code>	<code>ffffff</code>	Kartalar
 <code>--ink</code>	<code>#1a1410</code>	Asosiy matn (charcoal)
 <code>--accent</code>	<code>ff6b35</code>	Orange — YAGONA birlamchi urgʻu
 <code>--amber</code>	<code>ffb703</code>	Highlight, gradient sherigi
 <code>--success</code>	<code>#1fa855</code>	Faqat semantik (ochiq/tayyor) — urgʻu emas

Dark: naiv invert QILINMAYDI — kontrast qayta sozlangan (`--bg:#100d0b` , orange yumshatilgan `#ff7a47`).

Tipografiya

Inter (barcha rollar). Type scale `clamp` mobil→desktop.
Sarlavha: `text-wrap:balance` , `-0.02em` , weight 600–700.
Body 1.6, ~65ch. Raqamlar: tabular-nums.

Shakl va fazo

Radius: kartalar 20–24px, kontrol 12–14px, pill ∞.
Yumshoq koʻp-qatlamli soya (delivery «hard shadow» emas). Boʻlim padding saxiy (py-20→28), 8px grid.

SSR ishonchliligi (muhim qoida): kirish animatsiyalari sof CSS (`animate-fade-up`) bilan. Framer Motion `initial opacity:0` / `whileInView` kontentni SSR/preview'da yashiradi — ishlatilmaydi.

backend/ — umumiy API server PLAN

Node ≥20 + TypeScript + Fastify + zod. Mock data → keyin PostgreSQL/Prisma. Deploy: Railway (alohida service).

Endpointlar (reja)

METOD	YO'L	VAZIFA
POST	/api/chat	{text,lang,location} → Gemini niyat + geo → {reply, restaurants[]}
POST	/api/nearby	{location,filters} → masofa bo'yicha saralangan ro'yxat
POST	/api/voice	pipeline: audio → {transcript, reply, audioUrl}
WS	/api/voice/live	realtime: audio stream ↔ Gemini Live proxy
POST	/api/reserve	{restaurantId,time,people} → bron
POST	/api/order	{items[],address,payment} → zakaz (dastavka hamkor)
GET	/health	200 healthcheck

Voice layer — `VoiceEngine` interfeysi (`transcribe` / `speak` / `realtimeSession`). Primary: Gemini Live (16kHz PCM mono WS). Fallback: STT→Gemini→TTS. Model ID'lari bitta config'da (`models.ts`) — hard-code yo'q (Gemini tez yangilanadi, context7'dan tekshiriladi).

Bosqichlar: 1) skeleton+config+health → 2) geo+mock data → 3) Gemini chat/intent → 4) voice pipeline → 5) voice realtime → 6) bron+auth (TG initData HMAC).

tg-mini-app/ & pwa/ PLAN

Ikki iste'molchi kanali, bir xil backend, bir xil restoran obyekt shakli.

tg-mini-app — asosiy kanal

Nega: O'zbekistonda Telegram dominant — ilova shu yerda "yashaydi".

Stek: Vite + React 18 + TS · @tma.js/sdk-react v3 · @telegram-apps/telegram-ui · react-router HashRouter · react-i18next.

Native: BackButton, MainButton, themeParams, hapticFeedback, initData auth (server HMAC), `requestLocation()`. SSR yo'q — statik SPA (CDN/GitHub Pages).

pwa — o'rnatiladigan ilova

Nega: Telegram'i yo'q foydalanuvchilar + iOS/Android home-screen.

Stek: Next.js 15 (website reuse) · Serwist SW · `app/manifest.ts` · next-intl · next-themes.

PWA yadrosi: manifest + service worker (offline/cache) + installable. `navigator.geolocation`, `MediaRecorder` voice, tab-bar (Chat/Tavsiya/Profil), push (keyin, VAPID).

Umumiy restoran obyekti (barcha loyihalarda bir xil, website `mock.ts` etalon): `{ id, name, dish, km, rating, priceFrom, waitMin, ready }`. Mijozlar website'dan `mock.ts`, i18n `messages`, design tokenlar va `Logo / Card` primitivlarini qayta ishlatadi — noldan yozilmaydi.

Holat, roadmap va kuchli tomonlar

Roadmap

#	BOSQICH	MAQSAD	HOLAT
1	website	Birinchi taassurot, investor/hamkor	✅ Bajarildi
2	backend	Gemini AI, voice layer, geo-qidiruv	📄 Keyingi
3	tg-mini-app	Asosiy iste'molchi kanali (TG dominant)	📄 Reja
4	pwa	Brauzer/home-screen, offline	📄 Reja
5	B2B + POS	Restoran dashboard, real POS (iiko/Jowi/Poster)	📄 Keyin

✅ Kuchli tomonlar

Aniq strategik pozitsiya (data-moat, delivery emas). Puxta hujjatlashtirilgan (har loyihada CLAUDE.md + PLAN.md). Yagona dizayn tizimi + reuse rejasi. Kalit-xavfsizligi arxitekturada mustahkam. Real bozor insaytiga asoslangan.

⚠️ Xatarlar / keyingi qadamlar

4 loyihadan faqat 1 tasi qurilgan — asosiy mahsulot (backend + mijozlar) hali kod emas. Real restoran data + bron integratsiyasi (moat) hali yo'q — eng og'ir ish oldinda. Gemini free token limitlariga bog'liqlik. B2B onboarding — muvaffaqiyat kaliti.

Umumiy baho: Kuchli strategiya va professional poydevor. Marketing sayt sifatli qurilgan. Asl qiymat (real-time data + bron) hali qurilmagan — bu ham eng katta imkoniyat, ham eng katta risk. Keyingi hal qiluvchi qadam: **backend + tg-mini-app**.